

Question 1: What is a null hypothesis (H_0) and why is it important in hypothesis testing?

- The null hypothesis (H_0) states that there is no effect, no difference, or no relationship between variables.
- It is the starting assumption in hypothesis testing.
- It helps researchers test ideas using data instead of assumptions.
- Sample results are compared against H_0 to see if they happened by chance.
- If enough evidence is found, H_0 is rejected; otherwise, it is not rejected.

Question 2: What does the significance level (α) represent in hypothesis testing?

- The significance level (α) shows the allowed chance of making an error.
- It is the probability of rejecting a true null hypothesis.
- This error is called a Type I error.
- Common values of α are 0.05 (5%) and 0.01 (1%).
- It helps decide whether the result is statistically significant.

Question 3: Differentiate between Type I and Type II errors.

- Type I Error: Occurs when the null hypothesis is rejected even though it is true (a false positive).
- Type II Error: Occurs when the null hypothesis is not rejected even though it is false (a false negative).
- Type I error is controlled by the significance level (α).
- Type II error is related to the power of the test ($1 - \beta$).

Question 4: Explain the difference between a one-tailed and two-tailed test. Give an example of each.

- A **one-tailed test** checks for an effect in **only one direction** (either greater than or less than).
 - **Example:** Testing whether a new teaching method **improves** student scores compared to the old method.
- A **two-tailed test** checks for an effect in **both directions** (either increase or decrease).
 - **Example:** Testing whether a new teaching method results in **different** student scores (higher or lower) than the old method.

Question 5: A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At $\alpha = 0.05$, test the claim.

Step 1: State the hypotheses

- Null hypothesis (H_0): $\mu = 10$ minutes
- Alternative hypothesis (H_1): $\mu \neq 10$ minutes

Step 2: Choose the test

- Small sample ($n = 9$) and population standard deviation unknown
- Use a one-sample t-test (two-tailed)

Step 3: Given data

- Sample mean ($\bar{x}$) = 12
- Standard deviation (s) = 3
- Sample size (n) = 9
- Significance level (α) = 0.05

Step 4: Calculate the test statistic

$$\left[\frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{12 - 10}{3/\sqrt{9}} = \frac{2}{1} = 2 \right]$$

Step 5: Find the critical value

- Degrees of freedom (df) = $n - 1 = 8$
- Critical t value at $\alpha = 0.05$ (two-tailed) = ± 2.306

Step 6: Make the decision

- Calculated $t = 2$
- Since $|2| < 2.306$, do not reject H_0

Step 7: Conclusion

At the 5% significance level, there is no sufficient evidence to reject the company's claim. So, the average complaint resolution time can be considered 10 minutes.

Question 6: When should you use a Z-test instead of a t-test?

- Use a Z-test when the population standard deviation (σ) is known.
- Use a Z-test when the sample size is large ($n \geq 30$).
- The data should be approximately normally distributed (or large sample size by Central Limit Theorem).
- If the population standard deviation is unknown and the sample size is small, use a t-test instead.